

SAHIL BHOIR

Software Engineer | Backend and Cloud | Applied ML

Thane, Maharashtra | [Email](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Medium](#)

SUMMARY

Computer Engineering graduate building software across backend APIs, cloud workflows, data processing and user-facing interfaces. I have delivered projects that connect authenticated services, asynchronous processing, testing and practical review tools. My applied ML work adds careful evaluation and clear limits to that engineering base.

PROJECTS

Serverless Receipt Processor | [AWS Lambda](#) · [Cognito](#) · [S3](#) · [Step Functions](#) · [Textract](#) · [DynamoDB](#) | [Demo](#) · [GitHub](#)

- Built an authenticated serverless workflow that transformed receipt images into user-scoped, editable expense records using Cognito, S3, Textract, Lambda and DynamoDB.
- Separated upload and processing workloads through short-lived presigned URLs, EventBridge and Step Functions, keeping asynchronous OCR and parsing outside the client request path.
- Improved extraction reliability with five focused parser regressions for ambiguous totals and Indian currency and date formats, while retaining the original receipt and editable fields when OCR remained uncertain.

Cyberattack Tactics Triage | [Python](#) · [TF-IDF](#) · [MITRE ATT&CK](#) | [GitHub](#)

- Achieved 0.773 macro F1, 0.826 micro F1 and 0.720 exact match by training a multilabel Linear SVM on 16,955 cleaned Enterprise ATT&CK procedure examples spanning 15 tactics.
- Strengthened evaluation reliability by replacing a random split with source-grouped partitions, removing eight cross-partition near-duplicates and testing on 3,391 grouped examples.
- Built an analyst-review workflow with calibrated confidence, feature-level explanations and CSV batch processing; an unseen-technique stress test fell to 0.516 macro F1, making the boundary on unfamiliar behaviour explicit.

Real-Time Violence Event Detection | [YOLOv8](#) · [OpenCV](#) · [FastAPI](#) · [Streamlit](#) | [Demo](#) · [GitHub](#)

- Evaluated 17,289 frame observations across 35 reviewed clips and selected a 0.70 confidence threshold with five positive activation frames and three negative release frames.
- Designed an N-positive/K-negative state machine that converted noisy pretrained YOLOv8 frame predictions into persistent, reviewable events while suppressing duplicate alerts.
- Delivered an end-to-end monitoring workflow for video, webcam and RTSP inputs with FastAPI controls, Streamlit review, stored screenshots, runtime diagnostics and optional Telegram alerts; detector-level misses remained separate from temporal-filter performance.

EDUCATION

Vidyalankar Institute of Technology | B.E. Computer Engineering, AI and ML Minor | **CGPA: 9.43/10** | 2021-2025

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C/C++

Backend and APIs: FastAPI, Flask, REST APIs, WebSockets, authentication, automated testing

AWS and DevOps: Lambda, S3, DynamoDB, API Gateway, Step Functions, EventBridge, Docker, GitHub Actions

Data and ML: Pandas, NumPy, scikit-learn, PyTorch, OpenCV

CERTIFICATIONS AND WRITING

Certifications: AWS Certified Cloud Practitioner (CLF-C02) · IBM Full Stack Software Developer Professional Certificate

Writing: Articles on AI systems, agents, work and computational mythology - [Medium profile](#)